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1 Battery Technology for Electrical Engineers: A Self-Study Curriculum

Target duration: 8-10 weeks part-time (roughly 8-12 hours/week)

Prerequisites: Undergraduate EE (circuits, linear algebra, ODEs, basic control theory)

Goal: Fluency in battery literature and readiness to begin SIB simulation research

1.1 Part I: Foundations (Weeks 1-2)

1.1.1 Chapter 1: Electrochemistry for Engineers

Learning objective: Read a half-reaction and understand what's physically happening.

1.1 What is a battery? Galvanic vs electrolytic cells

1.2 Oxidation, reduction, and half-reactions

1.3 Anode, cathode, electrolyte, separator, current collector — anatomy of a cell

1.4 Standard electrode potentials and the electrochemical series

1.5 The Nernst equation and what it predicts

1.6 Activity vs concentration (and why the distinction matters later)

- 1.7 Faraday's laws of electrolysis — linking charge to mass
- 1.8 Gibbs free energy and cell voltage

Deliverable: Write out the half-reactions for a Li-ion cell (graphite/LCO) and a Na-ion cell (hard carbon/layered oxide). Calculate theoretical cell voltage from standard potentials.

Primary source: Atkins, Physical Chemistry, electrochemistry chapter. Supplement with Huggins, Advanced Batteries, Ch. 1-2.

1.1.2 Chapter 2: How a Battery Works in Operation

Learning objective: Explain, without equations, what happens inside a cell during charge and discharge.

- 2.1 Intercalation — the guest-host mechanism
- 2.2 Why intercalation hosts matter (layered, spinel, olivine, polyanionic structures)
- 2.3 The solid-electrolyte interphase (SEI): formation, function, consequences
- 2.4 The electric double layer and double-layer capacitance
- 2.5 Charge transfer kinetics at the electrode-electrolyte interface
- 2.6 Mass transport: diffusion, migration, convection
- 2.7 Kinetic vs transport limitations — which dominates when
- 2.8 Introduction to the Butler-Volmer equation (intuition only; derivation comes later)

Deliverable: Draw a labeled diagram of a Li-ion cell during discharge showing ion flow, electron flow, and where each loss mechanism occurs.

Primary source: Huggins, Advanced Batteries, Ch. 3-4. Reddy's Linden's Handbook of Batteries Ch. 1 for breadth.

1.2 Part II: The Language of Batteries (Week 3)

1.2.1 Chapter 3: Performance Metrics and Terminology

Learning objective: Read a commercial cell datasheet and interpret every number and curve.

- 3.1 Capacity (Ah) vs energy (Wh); specific vs volumetric
- 3.2 C-rate and what "1C" actually means
- 3.3 State of charge, depth of discharge, state of health
- 3.4 Open-circuit voltage vs terminal voltage; why OCV curve shape matters for BMS
- 3.5 Internal resistance and the three polarizations (ohmic, activation, concentration)
- 3.6 Coulombic, voltage, and energy efficiency
- 3.7 Cycle life vs calendar life; what counts as "a cycle"
- 3.8 Self-discharge
- 3.9 CC-CV charging protocol
- 3.10 Characterization tests: HPPC, GITT, PITT, EIS — what each measures and why

Deliverable: Download a CATL or Samsung cell datasheet and a HiNa SIB datasheet. Write a one-page interpretation of each, explaining every graph and specification in your own words.

Primary source: Battery University articles (free, online). Plett, Battery Management Systems Vol. 1, Ch. 1-2.

1.3 Part III: Cells and Chemistries (Week 4)

1.3.1 Chapter 4: Cell Construction

Learning objective: Know how a cell is physically built and why construction choices matter.

- 4.1 Form factors: cylindrical (18650, 21700, 4680), pouch, prismatic
 - 4.2 Electrode manufacturing: slurry, coating, calendaring, slitting
 - 4.3 Formation cycling and why first cycles differ
 - 4.4 Current collectors (Cu, Al) and why SIB can use Al on both sides
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1.3.2 Chapter 5: Lithium-Ion Chemistry Families

Learning objective: Name the major Li-ion chemistries and their tradeoffs without looking them up.

- 5.1 LCO, LFP, NMC, NCA, LMO, LTO — cathodes and anodes
 - 5.2 Energy, power, safety, cost, cycle life tradeoffs
 - 5.3 Which chemistry wins in which application (EV, grid, consumer, power tool)
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1.3.3 Chapter 6: Sodium-Ion Chemistry Families

Learning objective: Map the SIB landscape and identify which chemistries are commercialized.

- 6.1 Cathodes: layered oxides (O3, P2 types), polyanionic (NVPF, NFPP), Prussian blue analogues
- 6.2 Anodes: hard carbon (dominant), soft carbon, alloys, titanates
- 6.3 Electrolytes and why SIB electrolytes differ from Li-ion
- 6.4 Commercial cells: CATL, HiNa, Faradion/Reliance, Natron, Altris, Tiamat
- 6.5 Why Na⁺ is harder to work with than Li⁺ (larger ion, different host requirements)

Deliverable: Build a comparison table of five commercial SIB cells vs five commercial Li-ion cells across all Chapter 3 metrics.

Primary source: Reddy, Linden's Handbook of Batteries. For SIB: Hu Yong-Sheng review papers (search Google Scholar, read his two most-cited SIB reviews from 2018 onward).

1.4 Part IV: Why Batteries Die (Week 5)

1.4.1 Chapter 7: Degradation Mechanisms

Learning objective: Explain the physical origin of capacity fade and power fade in a specific cell under specific operating conditions.

- 7.1 The three modes: loss of lithium/sodium inventory, loss of active material, impedance growth
- 7.2 SEI growth — the dominant calendar aging mechanism
- 7.3 Lithium/sodium plating — when and why
- 7.4 Particle cracking and mechanical fatigue
- 7.5 Transition metal dissolution and crosstalk
- 7.6 Electrolyte decomposition and gas generation
- 7.7 Calendar aging vs cycle aging — different physics, different models
- 7.8 Stressors: temperature, SOC, DOD, C-rate, voltage limits

Deliverable: Read Birkel et al. (2017) “Degradation diagnostics for lithium ion cells” twice. Write a one-page summary in your own words identifying which mechanisms are diagnosable from external measurements alone.

Primary source: Birkel et al. 2017 review. Supplement with Vetter et al. 2005 — older but foundational.

1.5 Part V: Thermal Behavior (Week 6, first half)

1.5.1 Chapter 8: Heat in Batteries

Learning objective: Predict whether a cell will overheat under a given duty cycle.

- 8.1 Sources of heat generation: ohmic, polarization, entropic (reversible)
- 8.2 Bernardi’s equation for heat generation
- 8.3 Heat transfer out of a cell; thermal time constants
- 8.4 Safe operating windows and why they exist
- 8.5 Thermal runaway: triggers, stages, propagation
- 8.6 Why low temperature hurts performance and high temperature accelerates aging
- 8.7 Why SIB is safer than Li-ion (and how to quantify that)

Deliverable: Hand-calculate steady-state heat generation for a commercial cell at 1C discharge using datasheet internal resistance. Compare to a rough convective cooling estimate.

1.6 Part VI: Battery Systems and BMS (Weeks 6-8)

1.6.1 Chapter 9: Pack Architecture

- 9.1 Series and parallel configurations; nomenclature (e.g., 96s2p)
 - 9.2 Cell-to-cell variation and its consequences
 - 9.3 Contactors, precharge, fuses, current sensing
 - 9.4 CAN bus and BMS communication basics
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1.6.2 Chapter 10: State Estimation

Learning objective: Understand why SOC estimation is non-trivial and what each method trades off.

- 10.1 Coulomb counting and its drift problem
 - 10.2 OCV-based SOC lookup and why flat OCV curves (SIB hard carbon!) break it
 - 10.3 Model-based estimation: the ECM + Kalman filter approach
 - 10.4 SOH estimation: capacity vs resistance based
 - 10.5 Remaining useful life (RUL) prediction
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1.6.3 Chapter 11: Cell Balancing

- 11.1 Why cells drift; consequences of imbalance
 - 11.2 Passive balancing (resistive bleed)
 - 11.3 Active balancing topologies
 - 11.4 When to balance (top, bottom, or throughout)
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1.6.4 Chapter 12: Functional Safety (Brief)

- 12.1 ISO 26262 and ASIL ratings — awareness level only
- 12.2 BMS failure modes and protective functions

Deliverable: Complete Plett’s Coursera specialization “Algorithms for Battery Management Systems,” Courses 1 and 2 at minimum. Work through the MATLAB assignments even if loosely.

Primary source: Plett, Battery Management Systems Vol. 1 and Vol. 2. This is the EE-friendly canonical text — read it cover to cover during this part.

1.7 Part VII: Sodium-Ion Deep Dive (Weeks 9-10)

1.7.1 Chapter 13: What’s Different About Sodium

Learning objective: Identify every place existing Li-ion modeling and BMS methods need adaptation for SIB.

- 13.1 Na⁺ vs Li⁺: ionic radius, mass, standard potential, diffusion
 - 13.2 Hard carbon anode behavior: the slope region vs the plateau region
 - 13.3 Why the plateau breaks OCV-based SOC estimation
 - 13.4 Al current collectors on both sides — cost and implications
 - 13.5 Low-temperature advantage — physical origin
 - 13.6 Safety advantage — physical origin
 - 13.7 SEI differences in SIB
 - 13.8 Degradation modes unique to or amplified in SIB
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1.7.2 Chapter 14: The SIB Research Landscape

- 14.1 Major research groups (Hu Yong-Sheng, Tarascon, Passerini, Palacín, Adelhelm)
- 14.2 Chinese research dominance — where to look (CNKI, IOP-CAS, Chinese journals)
- 14.3 Commercial landscape update (who shipped what in 2025-2026)
- 14.4 Where the open problems are — and which are addressable by EE simulation

Deliverable: Read three recent (2024–2026) SIB review papers. Produce a personal “open questions” document listing every problem mentioned as unsolved, tagged by whether it’s a materials problem or a systems/modeling problem.

Primary source: Hu Yong-Sheng reviews. Recent Nature Energy and Joule review articles on SIB. Faraday Institution SIB reports if available.